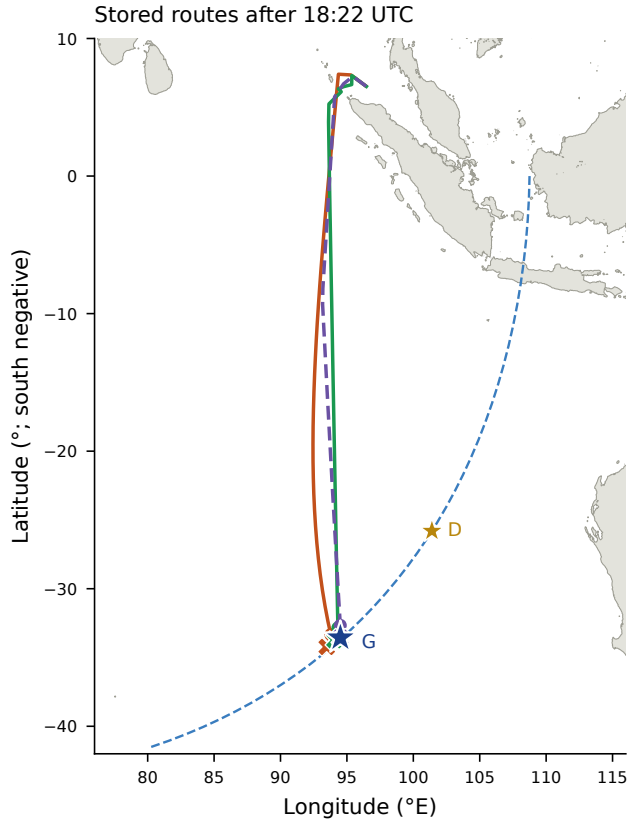
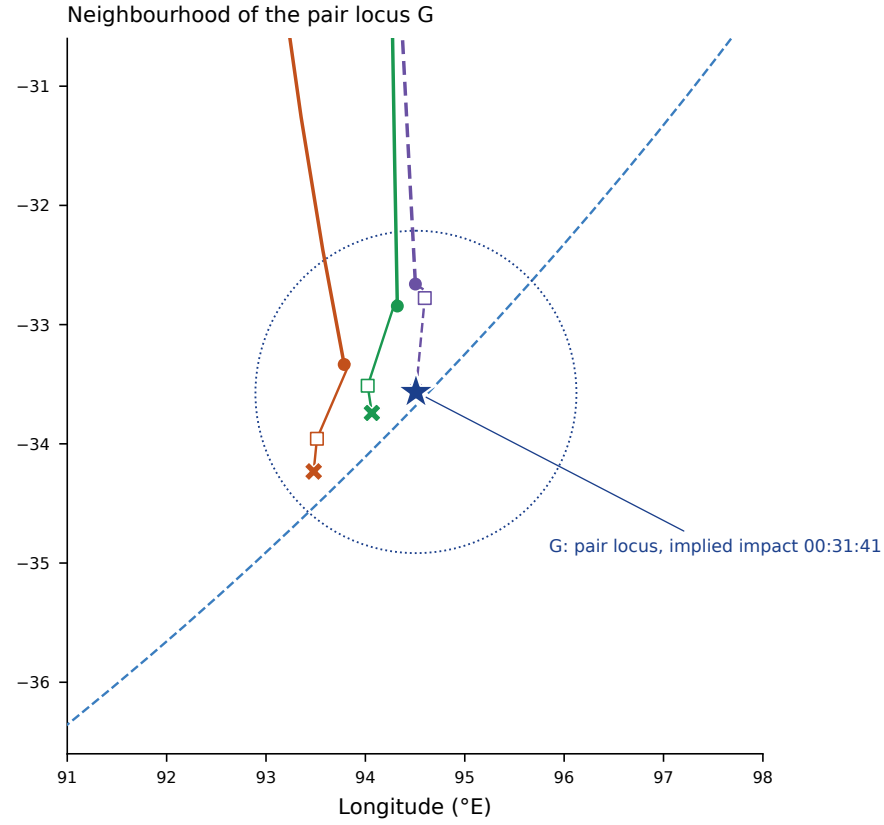
Near the pair locus the simulated impacts span 00:20-00:45 UTC (median 00:31:21); 18 % of the nearby weight falls within 2 min of the implied 00:31:41





highest weight within 150 km
highest weight with impact within 2 min of 00:31:41
closest to G (any non-zero weight)
route at 00:10:59
start of the final descent
simulated impact
150 km from G

highest weight within 150 km: 120 km from G; impact 00:20:57 (-11 min against 00:31:41); final descent from 00:16:45, max 40,722 ft/min; family B; weight share 2.7e-05

highest weight with impact within 2 min of 00:31:41: 45 km from G; impact 00:32:42 (+1 min against 00:31:41); final descent from 00:17:57, max 13,616 ft/min; family B; weight share 2.4e-05

closest to G (any non-zero weight): 0 km from G; impact 00:26:50 (-5 min against 00:31:41); final descent from 00:11:34, max 38,837 ft/min; family B; weight share 2.4e-19

(i) Three stored flight paths from the simulation are shown. Orange has the highest weight within 150 km of G. Green has the highest weight among those that also hit the sea within 2 minutes of the time the pair gives. Purple is the closest to G, but its weight is almost zero. The flight paths and the impacts are linked by position at 00:11, not by a direct record. These are stored samples; they were not fitted to the sounds. The two sounds (Cape Leeuwin 00:52:04, Diego Garcia 01:12:49) have a time difference that agrees with one point on the arc, 33.6°S 94.5°E. Pairs like this occur by chance: the test gives p 0.019 from 36 tests, which is a chance level. The Diego Garcia sound is not kept when the airgun noise is removed. The sound direction that Kadri published for 00:52 does not point to this place. The impact would need about 330 times more sound than the calibration crash gives.

(ii) Pair G: Cape Leeuwin H01W 00:52:04 UTC (DIGITISED Kadri 2024 Fig. 9 panel c; envelope matched filter M1b p 0.043; Kadri's 'text time 00:52:03', 'caption time 00:54:30') x Diego Garcia South H08S 01:12:49 (DIGITISED panel e, raw, M1b p 0.039; the airgun-suppressed residual does not keep it, p_B 0.30); lag 1,245 s inside the predicted 809-1,536 s; 19 lag-consistent pairs vs 8.8 expected, permutation p 0.019 over 36 receiver-pair tests (chance level: a smallest p this low arises about half the time). Arc locus 33.6°S 94.5°E, implied impact 00:31:41; eta for P_D(both) = 0.5 is 0.069 = 330x the F-35A anchor 2.1e-4. BEARING: the locus lies at 269.0° from H01W (1,812 km) and 144.8° from H08S (3,657 km), WGS84; Kadri gives 57° (text p. 9) or 306.18° (Table 1 caption) for the 00:52 signal; neither is within 37° of the locus bearing; nearest Table 1 row 00:52:50 at 274.84° (46 s later). Within 150 km of G: 8,550,080 end-of-flight rows (24 waypoints seeds), mass 0.045 of the D0' p 0.5 waypoints-stratum total; weighted impact-time quantiles 5/50/95 % 00:20:31 / 00:31:21 / 00:45:25; share with |t - 00:31:41| < 120 s = 0.181. Family shares within 150 km: A1 0.13, A2 0.28, B 0.59. Routes: core run J routes.npy (2,000 per seed, every 600 s from 18:01:49), joined to end-of-flight rows by POSITION at 00:10:59 (seed/route/distance: 11/1392/4.8 km; 8/466/5.4 km; 18/892/9.9 km); the routes carry no row identity. Weights: run_j_windows.ln_weights(D0') x p_factor(0.5). Source posterior run J, 00:19 Held Out (D0'), dive probability p 0.5; PREDICTIVE, NOT CALIBRATED; caveats C1 (propagation slope misfit, +10.4/+7.4 dB per octave at H01W/H08S) and C2 (coupling efficiency not calibrated; eta swept over four decades, F-35A upper anchor). Module branch hypothesis/hydroacoustics 9b27786; searches results-data/searches_run_j.